# Explanation, Features and Development of Personal Portfolio Website

## 1. Introduction

The given code is used to develop a Personal Portfolio Website for Hari Venkat, a first-year Computer Science and Engineering student specializing in Artificial Intelligence and Machine Learning (CSE-AIML).

The website is created using HTML and CSS. It presents the student's personal information, skills, projects, educational background, hobbies, achievements, resume, and contact details in an organized and professional manner.

The website uses a dark background with cyan highlights, making it attractive and easy to navigate. It is also designed to work on desktops, tablets, and mobile devices.

## 2. Explanation of the Code

The website is developed using two main technologies: HTML and CSS.

### A. HTML (HyperText Markup Language)

HTML is used to create the structure and content of the website. It defines headings, paragraphs, navigation links, images, buttons, and different sections.

The main HTML elements used in the code are:

| HTML element | Purpose |
| --- | --- |
| `<!DOCTYPE html>` | Declares the document as HTML5. |
| `<html>` | Defines the root of the HTML document. |
| `<head>` | Contains page settings, title, and stylesheet references. |

| | |
|---|---|
| `<title>` | Sets the title displayed in the browser tab. |
| `<body>` | Contains all visible website content. |
| `<header>` | Contains the website's navigation area. |
| `<nav>` | Provides links to different sections. |
| `<section>` | Organizes content into separate sections. |
| `<div>` | Groups elements for layout and styling. |
| `<h1>`, `<h2>`, `<h3>` | Display headings of different levels. |
| `<p>` | Displays text and descriptions. |
| `<a>` | Creates navigation links and buttons. |
| `<img>` | Displays the student's profile photograph. |
| `<footer>` | Displays copyright information at the bottom. |

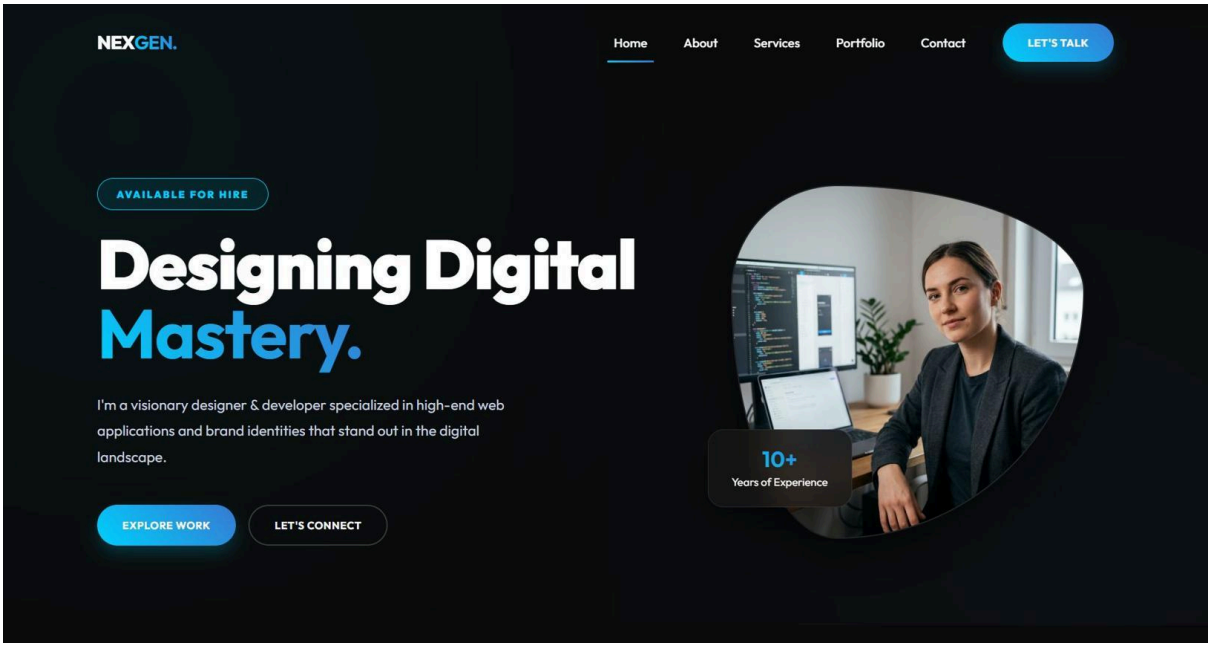
## B. CSS (Cascading Style Sheets)

CSS is used to control the appearance and layout of the website. It defines colors, fonts, spacing, borders, grids, alignment, and responsive behavior.

The important CSS properties used are:

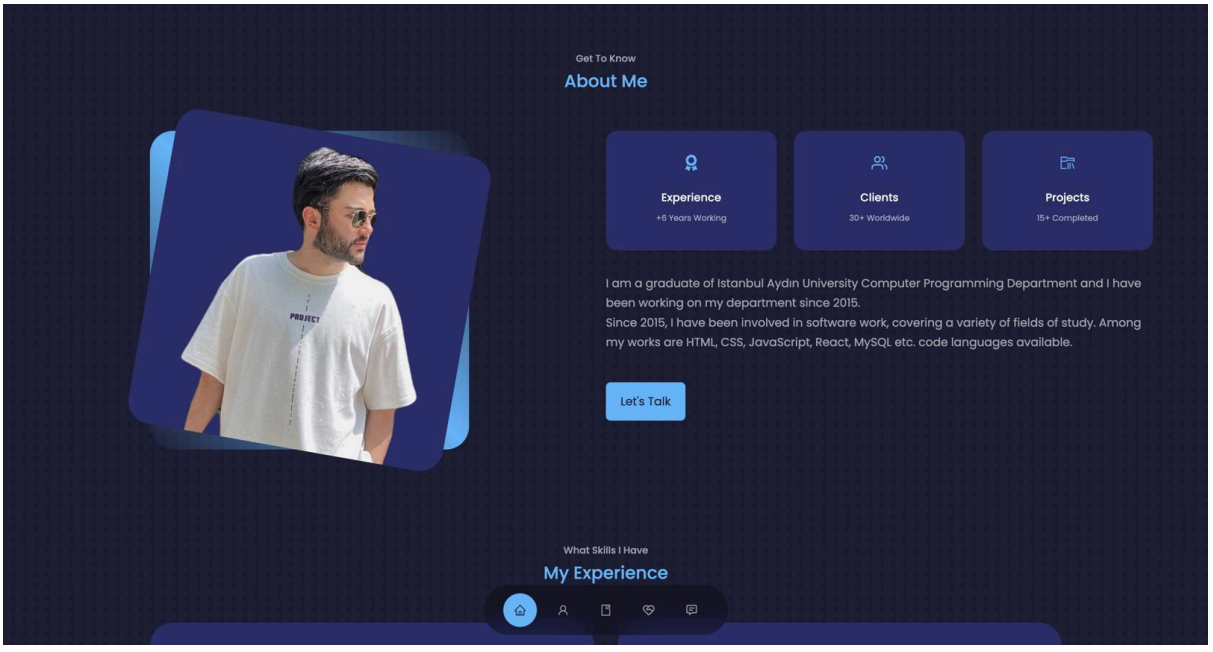
| CSS property | Purpose |
| --- | --- |
| `background-color` | Sets the background color. |
| `color` | Sets the text color. |
| `font-family` | Defines the font style. |
| `margin` | Adds space outside elements. |
| `padding` | Adds space inside elements. |
| `display: flex` | Aligns elements in flexible rows or columns. |
| `display: grid` | Arranges content in rows and columns. |
| `border` | Adds borders around elements. |
| `max-width` | Limits the maximum width of content. |
| `text-align` | Controls horizontal text alignment. |
| `@media` | Applies styles based on screen width. |
| `scroll-behavior: smooth` | Makes navigation scrolling smooth. |

# 3. Features of the Website



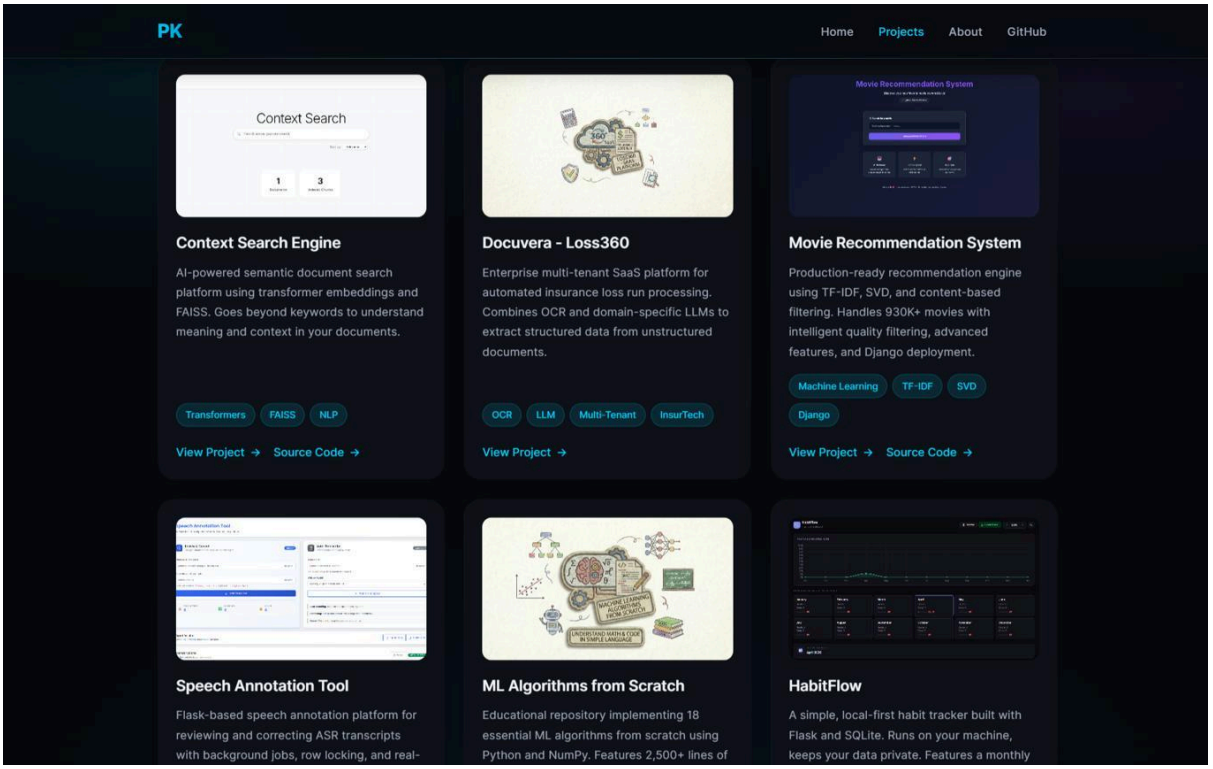

1. Home Section

Introduces Hari Venkat, displays his name, CSE-AIML specialization, a short introduction, profile photograph, and buttons to view projects or contact him.



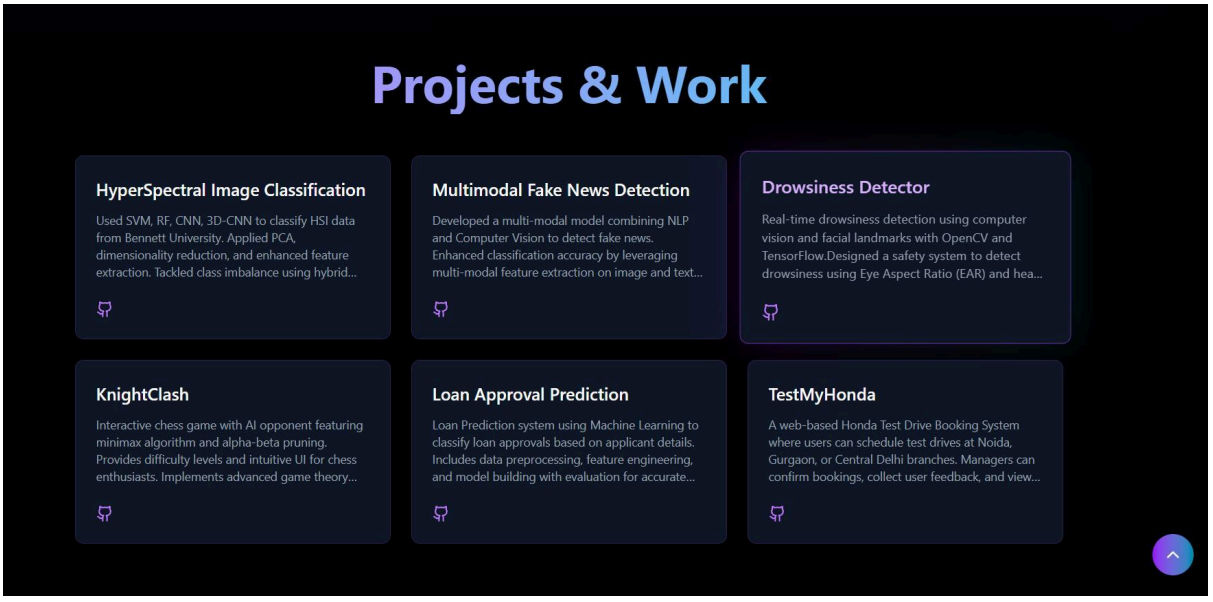

2. About Me Section

Describes the student's background, interests, programming experience, and learning journey. Information cards highlight the first year of study, learning duration, and skills.



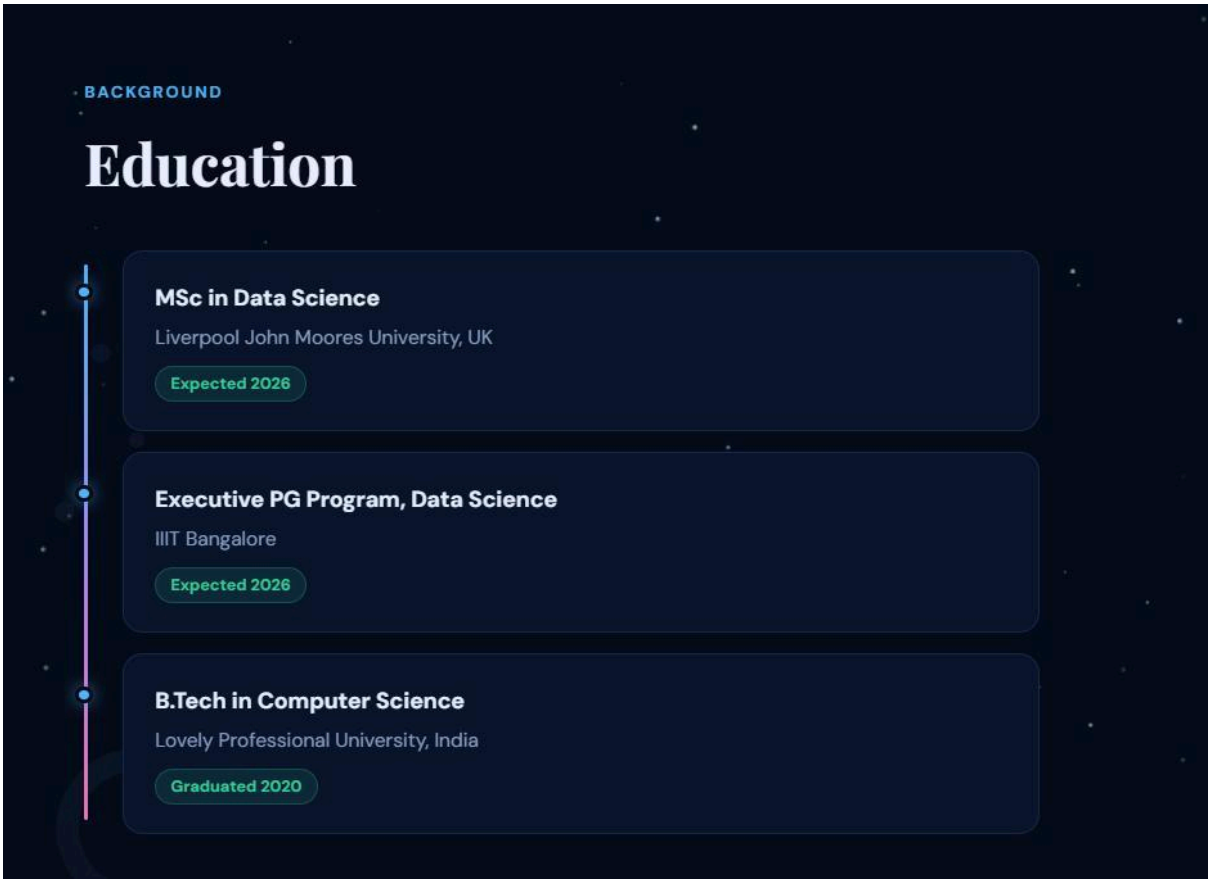

3. Skills Section

Displays Python, HTML, CSS, and AI/ML, along with short descriptions of the knowledge and concepts being learned.



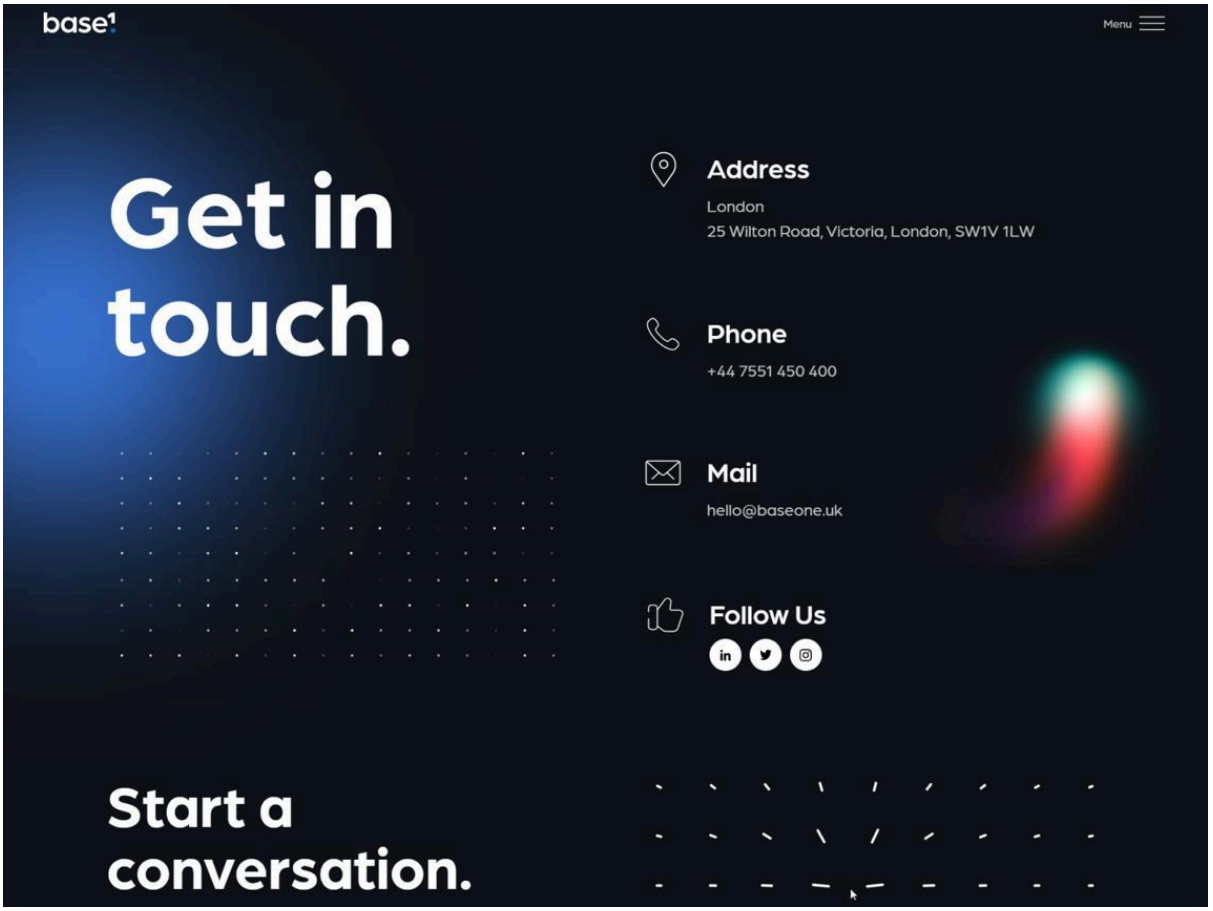

4. Projects Section

Presents three projects: Personal Portfolio, Online Learning Platform, and Python Programs. Each card includes a description and the technologies used.




5. Education Section

Displays the current B.Tech CSE (AI & ML) program and previous education details.




6. Contact Section

Displays email and location information, along with GitHub and LinkedIn links so visitors can connect with the student.

Other important features include:

- Education and achievements: Shows academic details, programming progress, and learning goals.
- Hobbies: Highlights web designing, learning technology, problem-solving, and exploring AI.
- Resume access: Provides a button to open a PDF resume.
- Navigation menu: Allows visitors to jump directly to different sections.
- Responsive layout: Adjusts the content arrangement for smaller screens.
- Consistent styling: Uses a dark theme with cyan highlights throughout the website.

# 4. Development of the Website

The website can be developed through the following steps.

Step 1: Create the HTML file

Create a file named `index.html`. Add the basic HTML structure and create the navigation, home, about, skills, projects, education, hobbies, achievements, resume, contact, and footer sections.

Step 2: Add CSS styling

Create a file named `style.css` and add styles for colors, typography, spacing, navigation, buttons, content cards, and page layouts.

Step 3: Design the sections

Use Flexbox for the home section and navigation, and CSS Grid for skills, projects, education, hobbies, and achievements.

Step 4: Make the website responsive

Use media queries at widths of 800px and 550px to rearrange columns, resize text and images, and improve the layout on tablets and mobile devices.

Step 5: Add personal files and links

Place `myphoto.jpg` and `resume.pdf` in the correct folder. Replace the sample email address with the actual contact email and verify the GitHub and LinkedIn links.

Step 6: Test the website

Open `index.html` in a browser. Check navigation links, image loading, resume access, social links, spacing, and mobile responsiveness.

# 5. Website Folder Structure

Keep the files organized as shown below.

Hari-Portfolio/

index.html

Website structure and content

style.css

Website design and styling

myphoto.jpg

Profile photograph

resume.pdf

Resume document

Important correction: Your code already includes a `<link rel="stylesheet" href="style.css">` reference, but the CSS is also written inside a `<style>` element in the HTML. For a cleaner structure, move all CSS into `style.css` and keep the stylesheet link inside `<head>`. Also, the HTML shown in your message contains escaped characters such as `\<html>`; in the actual file, use normal HTML tags such as `<html>`. The GitHub and LinkedIn `href` values should contain the direct URLs, not Markdown link syntax.

# 6. Technologies Used

| Technology | Usage |
| --- | --- |
| HTML5 | Creates the website structure. |
| CSS3 | Styles the page and controls its layout. |
| Flexbox | Aligns navigation and home content. |
| CSS Grid | Organizes cards into columns. |
| Media Queries | Adapts the layout to screen sizes. |
| Web Browser | Displays and runs the website. |

The website is a static front-end website. It does not currently use JavaScript, a database, or a server-side application. The contact information and social links are displayed as static content, and the resume button opens a PDF file if that file exists at the specified location.

# 7. Advantages of the Website

1. Presents the student's profile in a professional format.
2. Helps showcase technical skills and academic projects.
3. Provides easy navigation through different sections.
4. Offers access to the resume and social media profiles.
5. Uses a modern dark theme with consistent colors.
6. Supports desktop, tablet, and mobile screen sizes.

7. Helps the student build a professional online presence.

# 8. Conclusion

The Personal Portfolio Website is a simple and effective web development project created using HTML5 and CSS3. It combines personal information, technical skills, projects, education, hobbies, achievements, and contact details into a single website.

Developing this website helps demonstrate fundamental web development concepts such as semantic HTML, CSS styling, Flexbox, Grid, navigation links, and responsive design. It can also be extended in the future with JavaScript-based animations, a working contact form, project links, and additional interactive features.

# 9. Output

Hari.

Home About Skills Projects Education Contact

Hello, I'm

# Hari Venkat

CSE - AI & ML Student




I am a first-year CSE-AIML student interested in Python, Web Development and Artificial Intelligence.

My Projects | Contact Me

## About Me

I am a first-year Computer Science and Engineering student specializing in Artificial Intelligence and Machine Learning.

I have been learning Python and Web Development for around 4 months. I started with programming basics and gradually started creating simple websites using HTML and CSS.

I am interested in learning new technologies, improving my coding skills and understanding Artificial Intelligence.

01
First Year
CSE, AIML

04+
Months
Learning

04
Skills
Python • HTML • CSS • AI

## My Skills

**Python**
Python basics, loops, conditions, functions, lists, tuples and problem solving.

**HTML**
Creating webpages using headings, sections, links, tables, forms and semantic elements.

**CSS**
Working with colors, layouts, flexbox, responsive design and website styling.

**AI / ML**
Basic understanding of Artificial Intelligence, Machine Learning and Generative AI.

## My Projects

01
**Personal Portfolio**
A simple personal portfolio website created using HTML and CSS to display my profile, skills and projects.
HTML | CSS

02
**Online Learning Platform**
A simple educational website interface with course cards, instructors and learning content.
HTML | CSS

03
**Python Programs**
A collection of small Python programs created while learning programming fundamentals.
Python

## Education

2026 - Present
**B.Tech - CSE (AI & ML)**
Dr. RVR & NR University
Currently pursuing my first year of Computer Science and Engineering with specialization in Artificial Intelligence and Machine Learning.

Completed
**Higher Secondary Education**
Completed my previous education and started my journey into Computer Science and technology.

## My Hobbies

**Web Designing**
Creating and trying different website designs.

**Learning Technology**
Learning new programming concepts and tools.

**Problem Solving**
Practicing Python programs and logical problems.

**Exploring AI**
Learning about AI and Generative AI tools.

## Achievements & Goals

**Python Fundamentals**
Built a basic foundation in Python through regular practice.

**Web Development**
Started creating webpages using HTML and CSS.

**Learning AI**
Started exploring Artificial Intelligence and Generative AI.

**My Resume**
You can view my resume to know more about my education, skills and projects.
View Resume

## Contact Me

**Let's Connect**
I am always interested in learning new things and connecting with people who are interested in technology.

**Email:** yourmail@gmail.com

**Location:** Andhra Pradesh, India

GitHub
LinkedIn

© 2026 Hari Venkat | CSE-AIML Student

**10. Code**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

   <title>Hari Venkat | CSE-AIML Student</title>

   <link rel="stylesheet" href="style.css">
</head>
<style>
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    background-color: #101419;
    color: #eeeeee;
    line-height: 1.6;
}

/* Navigation */

header {
    background-color: #101419;
```

```
    border-bottom: 1px solid #29313a;
}

.navbar {
    max-width: 1100px;
    margin: auto;
    padding: 20px;

   display: flex;
    align-items: center;
    justify-content: space-between;
}

.logo {
    font-size: 28px;
}

.logo span {
    color: #65d5f5;
}

.navbar ul {
    display: flex;
    gap: 25px;
    list-style: none;
}

.navbar a {
    color: #cccccc;
    text-decoration: none;
    font-size: 14px;
}

.navbar a:hover {
    color: #65d5f5;
}
```

```
/* Home */

.home {
    max-width: 1100px;
    min-height: 90vh;
    margin: auto;
    padding: 70px 20px;

   display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 60px;
}

.home-text {
    max-width: 600px;
}

.home-text > p:first-child {
    color: #65d5f5;
    font-size: 14px;
    margin-bottom: 10px;
}

.home-text h1 {
    font-size: 65px;
    margin-bottom: 10px;
}

.home-text h2 {
    font-size: 25px;
    color: #c5cbd1;
    font-weight: normal;
    margin-bottom: 20px;
}
```

```
.home-text p {
    color: #9da5ad;
    max-width: 550px;
}

.buttons {
    display: flex;
    gap: 15px;
    margin-top: 30px;
}

.buttons a {
    text-decoration: none;
    padding: 11px 20px;
    border: 1px solid #65d5f5;
    color: #65d5f5;
}

.buttons a:first-child {
    background-color: #65d5f5;
    color: #101419;
}

.home-photo {
    width: 330px;
}

.home-photo img {
    width: 330px;
    height: 390px;
    object-fit: cover;
    border: 1px solid #39434d;
    padding: 8px;
}
```

```
/* Common Sections */

.section {
    max-width: 1100px;
    margin: auto;
    padding: 90px 20px;
}

.title {
    font-size: 38px;
    margin-bottom: 45px;
}

.title::after {
    content: "";
    display: block;
    width: 45px;
    height: 3px;
    background-color: #65d5f5;
    margin-top: 10px;
}

/* About */

.about {
    display: grid;
    grid-template-columns: 1.5fr 1fr;
    gap: 60px;
}

.about p {
    color: #9da5ad;
    margin-bottom: 18px;
}
```

```
.about-info {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 15px;
}

.about-info div {
    background-color: #171d23;
    border: 1px solid #29313a;
    padding: 25px;
}

.about-info h3 {
    color: #65d5f5;
    font-size: 25px;
}

.about-info p {
    color: white;
    margin: 5px 0;
}

.about-info small {
    color: #7f8992;
}

/* Skills */

.skills {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 20px;
}

.skill {
    background-color: #171d23;
```

```
    border: 1px solid #29313a;
    padding: 25px;
}

.skill h3 {
    color: #65d5f5;
    margin-bottom: 10px;
}

.skill p {
    color: #929ba4;
    font-size: 14px;
}

/* Projects */

.projects {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 20px;
}

.project {
    background-color: #171d23;
    border: 1px solid #29313a;
    padding: 25px;
    min-height: 280px;
}

.project > span {
    color: #65d5f5;
    font-size: 13px;
}
```

```
.project h3 {
    margin: 35px 0 15px;
}

.project p {
    color: #929ba4;
    font-size: 14px;
}

.project .technology {
    color: #65d5f5;
    margin-top: 20px;
}

/* Education */

.education {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 25px;
}

.education-box {
    background-color: #171d23;
    border-left: 3px solid #65d5f5;
    padding: 30px;
}

.year {
    color: #65d5f5;
    font-size: 13px;
}

.education-box h3 {
    margin: 12px 0 5px;
}
```

```
.education-box h4 {
    color: #c5cbd1;
    font-weight: normal;
    margin-bottom: 15px;
}

.education-box p:last-child {
    color: #929ba4;
}

/* Hobbies */

.hobbies {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 15px;
}

.hobbies div {
    background-color: #171d23;
    border: 1px solid #29313a;
    padding: 25px;
}

.hobbies h3 {
    color: #65d5f5;
    margin-bottom: 10px;
}

.hobbies p {
    color: #929ba4;
    font-size: 14px;
}

/* Achievements */
```

```
.achievements {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 20px;
}

.achievements div {
    background-color: #171d23;
    border: 1px solid #29313a;
    padding: 25px;
}

.achievements h3 {
    margin-bottom: 10px;
}

.achievements p {
    color: #929ba4;
    font-size: 14px;
}

/* Resume */

.resume {
    max-width: 1060px;
    margin: 20px auto 90px;
    padding: 40px;

   background-color: #171d23;
    border: 1px solid #29313a;

   display: flex;
    justify-content: space-between;
    align-items: center;
}
```

```
.resume p {
    color: #929ba4;
    margin-top: 8px;
}

.resume a {
    background-color: #65d5f5;
    color: #101419;
    padding: 12px 22px;
    text-decoration: none;
}

/* Contact */

.contact {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 70px;
}

.contact-text h3 {
    font-size: 28px;
    margin-bottom: 15px;
}

.contact-text p {
    color: #929ba4;
    margin-bottom: 15px;
}

.social {
    display: flex;
    flex-direction: column;
}
```

```
.social a {
    color: #eeeeee;
    text-decoration: none;
    padding: 18px 0;
    border-bottom: 1px solid #29313a;
}

.social a:hover {
    color: #65d5f5;
}

/* Footer */

footer {
    border-top: 1px solid #29313a;
    text-align: center;
    padding: 25px;
    color: #727c85;
    font-size: 13px;
}

/* Responsive */

@media (max-width: 800px) {

   .navbar {
        flex-direction: column;
        gap: 15px;
    }

   .navbar ul {
        flex-wrap: wrap;
        justify-content: center;
    }
```

```
.home {
    flex-direction: column-reverse;
    text-align: center;
}

.home-text {
    max-width: 600px;
}

.buttons {
    justify-content: center;
}

.about,
 .contact {
    grid-template-columns: 1fr;
}

.projects {
    grid-template-columns: 1fr 1fr;
}

.hobbies {
    grid-template-columns: 1fr 1fr;
}

.achievements {
    grid-template-columns: 1fr;
}

.resume {
    margin-left: 20px;
    margin-right: 20px;
    flex-direction: column;
    text-align: center;
    gap: 25px;
```

```
    }
}

@media (max-width: 550px) {

    .home-text h1 {
        font-size: 45px;
    }

    .home-text h2 {
        font-size: 20px;
    }

    .home-photo,
     .home-photo img {
        width: 260px;
    }

    .home-photo img {
        height: 310px;
    }

    .skills,
     .projects,
     .education,
     .hobbies,
     .about-info {
        grid-template-columns: 1fr;
    }

    .section {
        padding: 65px 20px;
    }

    .title {
        font-size: 32px;
```

```
    }
}
</style>
<body>

  <!-- Navigation -->
   <header>
      <nav class="navbar">

        <h2 class="logo">Hari<span>.</span></h2>

        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#skills">Skills</a></li>
            <li><a href="#projects">Projects</a></li>
            <li><a href="#education">Education</a></li>
            <li><a href="#contact">Contact</a></li>
         </ul>

      </nav>
   </header>

  <!-- Home -->
   <section class="home" id="home">

     <div class="home-text">

        <p>Hello, I'm</p>

        <h1>Hari Venkat</h1>

        <h2>CSE - AI & ML Student</h2>

        <p>
            I am a first-year CSE-AIML student interested in
```

```
            Python, Web Development and Artificial Intelligence.
          </p>

          <div class="buttons">
            <a href="#projects">My Projects</a>
            <a href="#contact">Contact Me</a>
          </div>

        </div>

        <div class="home-photo">

          <img src="myphoto.jpg" alt="Hari Venkat">

        </div>

    </section>

    <!-- About -->
     <section class="section" id="about">

        <h2 class="title">About Me</h2>

        <div class="about">

          <div>
            <p>
              I am a first-year Computer Science and
Engineering
              student specializing in Artificial Intelligence and
              Machine Learning.
            </p>

            <p>
              I have been learning Python and Web
Development
              for around 4 months. I started with programming
```

```
            basics and gradually started creating simple
            websites using HTML and CSS.
        </p>

        <p>
            I am interested in learning new technologies,
            improving my coding skills and understanding
            Artificial Intelligence.
        </p>
    </div>

    <div class="about-info">

        <div>
            <h3>01</h3>
            <p>First Year</p>
            <small>CSE-AIML</small>
        </div>

        <div>
            <h3>04+</h3>
            <p>Months</p>
            <small>Learning</small>
        </div>

        <div>
            <h3>04</h3>
            <p>Skills</p>
            <small>Python • HTML • CSS • AI</small>
        </div>

    </div>

  </div>

</section>
```

```
<!-- Skills -->
 <section class="section" id="skills">

    <h2 class="title">My Skills</h2>

    <div class="skills">

        <div class="skill">
            <h3>Python</h3>
            <p>
                Python basics, loops, conditions, functions,
                lists, tuples and problem solving.
            </p>
         </div>

        <div class="skill">
            <h3>HTML</h3>
            <p>
                Creating webpages using headings, sections,
                links, tables, forms and semantic elements.
            </p>
         </div>

        <div class="skill">
            <h3>CSS</h3>
            <p>
                Working with colors, layouts, flexbox,
                responsive design and website styling.
            </p>
         </div>

        <div class="skill">
            <h3>AI / ML</h3>
            <p>
                Basic understanding of Artificial Intelligence,
```

```
            Machine Learning and Generative AI.
          </p>
        </div>

    </div>

</section>

<!-- Projects -->
 <section class="section" id="projects">

    <h2 class="title">My Projects</h2>

    <div class="projects">

        <div class="project">

            <span>01</span>

            <h3>Personal Portfolio</h3>

            <p>
                A simple personal portfolio website created
                using HTML and CSS to display my profile,
                skills and projects.
             </p>

            <p class="technology">
                HTML | CSS
             </p>

        </div>

        <div class="project">

            <span>02</span>

            <h3>Online Learning Platform</h3>
```

```
            <p>
                A simple educational website interface with
                course cards, instructors and learning content.
             </p>

            <p class="technology">
                HTML | CSS
             </p>

        </div>

        <div class="project">

            <span>03</span>

            <h3>Python Programs</h3>

            <p>
                A collection of small Python programs created
                while learning programming fundamentals.
             </p>

            <p class="technology">
                Python
             </p>

        </div>

    </div>

</section>

<!-- Education -->
 <section class="section" id="education">

    <h2 class="title">Education</h2>

    <div class="education">
```

```
        <div class="education-box">
            <p class="year">2026 - Present</p>
            <h3>B.Tech - CSE (AI & ML)</h3>
            <h4>Dr. RVR & NRI University</h4>
            <p>
                Currently pursuing my first year of
                Computer Science and Engineering with
                specialization in Artificial Intelligence
                and Machine Learning.
             </p>
        </div>
        <div class="education-box">
            <p class="year">Completed</p>
            <h3>Higher Secondary Education</h3>
            <p>
                Completed my previous education and
                started my journey into Computer Science
                and technology.
             </p>
        </div>
    </div>
</section>
<!-- Hobbies -->
 <section class="section">
    <h2 class="title">My Hobbies</h2>
```

```
      <div class="hobbies">

        <div>
          <h3>Web Designing</h3>
          <p>Creating and trying different website designs.</p>
        </div>

        <div>
          <h3>Learning Technology</h3>
          <p>Learning new programming concepts and
tools.</p>
        </div>

        <div>
          <h3>Problem Solving</h3>
          <p>Practicing Python programs and logical
problems.</p>
        </div>

        <div>
          <h3>Exploring AI</h3>
          <p>Learning about AI and Generative AI tools.</p>
        </div>

      </div>

    </section>

    <!-- Achievements -->
     <section class="section">

      <h2 class="title">Achievements & Goals</h2>

      <div class="achievements">

        <div>
          <h3>Python Fundamentals</h3>
```

```
            <p>
                Built a basic foundation in Python through
                regular practice.
            </p>
        </div>

        <div>
            <h3>Web Development</h3>
            <p>
                Started creating webpages using HTML and CSS.
            </p>
        </div>

        <div>
            <h3>Learning AI</h3>
            <p>
                Started exploring Artificial Intelligence
                and Generative AI.
            </p>
        </div>

    </div>

</section>

<!-- Resume -->
 <section class="resume">

    <div>
        <h2>My Resume</h2>

        <p>
            You can view my resume to know more about
            my education, skills and projects.
        </p>
    </div>
```

```
    <a href="resume.pdf" target="_blank">
        View Resume
     </a>

</section>

<!-- Contact -->
 <section class="section" id="contact">

    <h2 class="title">Contact Me</h2>

    <div class="contact">

        <div class="contact-text">

            <h3>Let's Connect</h3>

            <p>
                I am always interested in learning new
                things and connecting with people who are
                interested in technology.
             </p>

            <p>
                <strong>Email:</strong>
                yourmail@gmail.com
             </p>

            <p>
                <strong>Location:</strong>
                Andhra Pradesh, India
             </p>

        </div>

        <div class="social">
```

```
            <a href="https://github.com/Harivenkat-cpu"
target="_blank">
                GitHub
            </a>

            <a
href="https://www.linkedin.com/in/hari-venkat-512079412?utm_
source=share_via&utm_content=profile&utm_medium=member
_ios" target="_blank">
                LinkedIn
            </a>


        </div>

    </div>

  </section>

  <!-- Footer -->
   <footer>

    <p>
        © 2026 Hari Venkat | CSE-AIML Student
     </p>

  </footer>

</body>
</html>
```